Polynomial Inverses and Algebraic Integration

Counterexamples to Chen's Algebraic-Stability Conjecture
and a Complete Criterion on Fixed Polynomial Covers

Keio*

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Abstract

Chen's 2022 algebraic-stability conjecture proposes that algebraic functions admitting arbitrarily many algebraic primitives are precisely finite polynomial linear combinations of rational powers of shifted variables. We give counterexamples in every degree $d \geq 3$: an inverse branch of $t^d + t = x$ has an infinite primitive tower in $\mathbb{C}[t]$, yet generates a non-Galois extension of $\mathbb{C}(x)$ and therefore cannot belong to the proposed class. More generally, every polynomial inverse is algebraically stable, and it lies in that class exactly when its defining polynomial has the form $a(t-b)^d + c$. For a fixed polynomial cover $x = P(t)$, we classify all algebraically stable elements of $\mathbb{C}(t)$ as $\bigcup_{n\geq 0} \delta^n \mathbb{C}[t]$, where $\delta = P'(t)^{-1} d/dt$. Finite pole orders give an explicit bound reducing membership to one polynomial linear system. The proofs use classical substitution, field trace, valuations, and Galois theory. They refute the stated 2022 conjecture, but do not solve the full elementary-function classification in Chen–Kauers Problem 8. Historical priority is not certified. Two exact verification programs are included in the single TeX source.

1 The historical target and the scope of the result

Original Problem 8 of Chen–Kauers [1] asks which elementary functions remain elementary after arbitrarily many integrations. This is a broad classification problem. In his later study, Chen [2, Section 5] proposed a more specific description for functions whose repeated primitives can all be chosen algebraic. The counterexamples below concern that exact description. They are not a redefinition of the original elementary-function problem.

We distinguish two properties. A function is *algebraically stable* if it admits a coherent infinite sequence of primitives in $\overline{\mathbb{C}(x)}$. It is *elementary-stable* if those primitives may use elementary extensions, including logarithms. Algebraic stability implies elementary stability; the converse fails even on rationally parametrized algebraic curves. The distinction also separates both notions from stability of the minimum order of a differential equation, studied in [3].

Source verification and priority. The conjecture used here occurs in the conclusion of arXiv:2202.06305v1, printed page 14, and is retained in the author's eight-page conference-format manuscript, Section 5, PDF page 7. Its coefficient and exponent restrictions are reproduced mathematically below. A 2025 paper continues to study the related algebraic-stability problem through special fields [4, Proposition 5.3]. A search during preparation did not identify an earlier identical disproof. This limited search does not establish firstness, and the methods themselves are classical. The claims are the displayed theorems and the resulting contradiction to the stated conjecture, not a new foundational principle or an independently certified priority claim.

*Research development, independent internal checks, and manuscript preparation with ChatGPT.

2 Polynomial inverses and Chen's algebraic-stability conjecture

We address the specific algebraic-stability conjecture proposed in Section 5 of Chen's *Stability Problems in Symbolic Integration* [2]. Its asserted description of the elements stable under d/dx in $\overline{\mathbb{C}(x)}$ is the class of finite sums

$$\sum_{i=1}^{r} p_i(x)(x-c_i)^{\alpha_i}, \qquad p_i \in \mathbb{C}[x], \quad c_i \in \mathbb{C}, \quad \alpha_i \in \mathbb{Q} \setminus \{-1,-2,\dots\}. \tag{1}$$

Here stability means the existence of a sequence $f_0, f_1, \dots$ in the algebraic closure with $f_0 = f$ and $Df_{n+1} = f_n$ for every $n \geq 0$. We give counterexamples to the proposed necessity in (1). This is a statement about that specific conjecture; it is not a classification of all elementary-stable functions and does not close Chen–Kauers Problem 8. No assertion of historical priority is needed for the mathematical arguments below.

Proposition 2.1 (Polynomial pullback primitives)**.** *Let $P \in \mathbb{C}[t]$ be nonconstant and identify $x = P(t)$. Every $Q(t) \in \mathbb{C}[t]$ is stable under $D = d/dx$ in the finite algebraic extension $\mathbb{C}(t)/\mathbb{C}(x)$. In particular, every local inverse branch $t = t(x)$ of a polynomial admits an infinite tower of algebraic primitives in the same function field.*

Proof. The derivation extends uniquely to the function field, where

$$D = \frac{1}{P'(t)}\frac{d}{dt}.$$

Define a linear map on polynomials by

$$(J_P Q)(t) = \int_0^t P'(u)Q(u)\,du. \tag{2}$$

The integral in (2) is ordinary polynomial antidifferentiation, so $J_P Q \in \mathbb{C}[t]$, and the chain rule gives $D(J_P Q) = Q$. Consequently $f_n = J_P^n Q$ is the required infinite tower. Although D need not preserve the polynomial subring, the right-inverse construction does; no assertion that $\mathbb{C}[t]$ is a differential subring is being made. □

Proposition 2.2 (An abelian-extension obstruction)**.** *Every function of the form (1) generates a Galois extension of $\mathbb{C}(x)$ whose Galois group is abelian.*

Proof. Write $\alpha_i = a_i/m_i$ with $a_i \in \mathbb{Z}$ and $m_i \geq 1$, and choose u_i with $u_i^{m_i} = x - c_i$. Because $\mathbb{C}$ contains all roots of unity, $\mathbb{C}(x,u_i)/\mathbb{C}(x)$ is a cyclic Galois extension, allowing the trivial case. Its splitting polynomial is $T^{m_i} - (x - c_i)$. The compositum

$$E = \mathbb{C}(x, u_1, \dots, u_r)$$

is finite Galois, and its Galois group embeds into a product of cyclic groups, so is abelian. The displayed sum belongs to E. Every subgroup of an abelian group is normal; therefore every intermediate field, including the field generated by the sum, is Galois over $\mathbb{C}(x)$, with abelian group. □

Theorem 2.3 (Which polynomial inverses have the proposed form)**.** *Let $P \in \mathbb{C}[t]$ have degree $d \geq 1$, and identify $x = P(t)$. The function t has a representation of the form (1) if and only if*

$$P(t) = a(t-b)^d + c \qquad (a \in \mathbb{C}^*,\ b, c \in \mathbb{C}). \tag{3}$$

Proof. The sufficiency follows from $t = b + a^{-1/d}(x-c)^{1/d}$. Conversely, a representation of the proposed form implies by Proposition 2.2 that $\mathbb{C}(t)/\mathbb{C}(P(t))$ is Galois. It has degree d, so its automorphism group G has order d.

For $\sigma \in G$, write $\sigma(t) = r_\sigma(t) \in \mathbb{C}(t)$. The identity $P(r_\sigma(t)) = P(t)$ shows that r_σ has no finite pole and that $\deg r_\sigma = 1$. Hence the maps r_σ form a finite group of affine transformations. The point

$$b = \frac{1}{|G|} \sum_{\sigma \in G} r_\sigma(0)$$

is fixed by every map r_σ: applying any affine group element commutes with this average and permutes the terms. In the coordinate $u = t - b$, the group therefore consists of multiplications $u \mapsto \lambda u$. It is a subgroup of $\mathbb{C}^*$ of order d, hence the group of dth roots of unity. Invariance of the degree-d polynomial $P(b+u)$ then forces all its nonconstant monomials to have degrees divisible by d. Consequently $P(b+u) = au^d + c$ with $a \neq 0$, proving (3). □

Theorem 2.4 (Counterexamples in every degree at least three)**.** *For any integer $d \geq 3$, let t_d satisfy*

$$t_d^d + t_d = x. \tag{4}$$

Then t_d is stable under d/dx in $\overline{\mathbb{C}(x)}$, but it has no representation of the form (1). Hence the proposed necessity in Chen's algebraic-stability conjecture is false.

Proof. Stability follows from Proposition 2.1 with $P(t) = t^d + t$ and $Q(t) = t$. It remains to prove that $\mathbb{C}(t)/\mathbb{C}(t^d + t)$ is not Galois.

The polynomial $T^d + T - x$ is irreducible over $\mathbb{C}(x)$. Indeed $x - T^d - T$ is irreducible in $\mathbb{C}[x, T]$: viewed as a polynomial in x over $\mathbb{C}[T]$, its leading coefficient is a unit and its degree is one. Gauss's lemma then proves the asserted irreducibility over $\mathbb{C}(x)$. Thus the extension has degree d and is separable.

Suppose an automorphism over $\mathbb{C}(t^d + t)$ sends t to $r(t) \in \mathbb{C}(t)$. Then

$$r(t)^d + r(t) = t^d + t. \tag{5}$$

A finite pole of r would give a finite pole on the left, with the highest pole order coming from r^d and therefore unable to cancel. The right side has no finite poles, so r is a polynomial. Comparing degrees in (5) gives $\deg r = 1$; write $r(t) = at + b$ with $a \neq 0$. Since $d \geq 3$, comparison of the coefficient of t^{d-1} gives $da^{d-1}b = 0$, hence $b = 0$. Leading coefficients give $a^d = 1$, while the coefficient of t gives $a = 1$. Thus the identity is the only automorphism. An extension of degree $d > 1$ with only one automorphism is not Galois.

If t_d had the proposed representation, Proposition 2.2 would make its generated extension Galois, a contradiction. □

The same obstruction excludes even finite sums of rational multiples of rational-function radicals whose compositum is abelian. The proof uses polynomial substitution and elementary Galois theory; it does not introduce a new foundational principle or rely on a conjectural classification.

The cubic example and its first three primitives. For $x = t^3 + t$, choose all integration constants to make the polynomials vanish at $t = 0$. The first three primitives of t are

$$\begin{aligned} F_1(t) &= \frac{3}{4}t^4 + \frac{1}{2}t^2, \\ F_2(t) &= \frac{9}{28}t^7 + \frac{9}{20}t^5 + \frac{1}{6}t^3, \\ F_3(t) &= \frac{27}{280}t^{10} + \frac{117}{560}t^8 + \frac{19}{120}t^6 + \frac{1}{24}t^4. \end{aligned}$$

They satisfy $DF_1 = t$, $DF_2 = F_1$, and $DF_3 = F_2$, with $D = (3t^2+1)^{-1}d/dt$. An independent check of non-Galoisness in degree three is the discriminant

$$\operatorname{disc}_T(T^3 + T - x) = -4 - 27x^2,$$

which has two simple zeros and is not a square in $\mathbb{C}(x)$.

More generally the tower for (4) can be computed using the monomial rule

$$J_P(t^m) = \frac{d}{m+d}t^{m+d} + \frac{1}{m+1}t^{m+1}.$$

Starting from t, its nth member has degree $nd+1$. This exact recurrence also gives reproducible symbolic verification of as many initial levels as desired; such checks illustrate the uniform proof rather than replace it.

3 Complete algebraic-stability classification on a polynomial cover

Fix a nonconstant $P \in \mathbb{C}[t]$, put $x = P(t)$, and write $K = \mathbb{C}(t)$ and $\delta = P'(t)^{-1}d/dt$. The next result classifies all rational functions in this presentation, with algebraic primitives allowed in the entire algebraic closure. It is more restrictive than elementary integration, where logarithms are permitted, and more general than considering only the inverse function t itself.

Lemma 3.1 (Descent of algebraic primitives)**.** *Let K be a characteristic-zero differential field. If $f \in K$ and $\delta^n g = f$ for an element g algebraic over K, then there is $h \in K$ with $\delta^n h = f$.*

Proof. The finite extension $L = K(g)$ is stable under the uniquely extended derivation: differentiating the minimal polynomial expresses δg in L, since its derivative with respect to its indeterminate is nonzero at g. Every K-embedding of L into an algebraic closure commutes with the extended derivation, by uniqueness. Thus field trace commutes with δ and its powers. The normalized trace $h = [L:K]^{-1}\operatorname{Tr}_{L/K}(g)$ satisfies $\delta^n h = f$. □

Lemma 3.2 (Finite pole growth)**.** *At a finite point a, put $e_a = \operatorname{ord}_{t=a}(P(t) - P(a)) \geq 1$. If $g \in K$ has a pole of order $k > 0$ at a, then $\delta^n g$ has a pole of exactly $k + ne_a$ at a, for every $n \geq 0$.*

Proof. With $u = t - a$, write $P(t) - P(a) = cu^{e_a} + O(u^{e_a+1})$ and $g = bu^{-k} + O(u^{1-k})$, with $bc \neq 0$. Then

$$\delta g = -\frac{kb}{e_a c}u^{-k-e_a} + O(u^{1-k-e_a}).$$

The leading coefficient is nonzero in characteristic zero. Iterating the same calculation proves the assertion. □

Theorem 3.3 (Classification and a finite decision bound)**.** *For nonzero $f \in \mathbb{C}(t)$, let m_a be its pole order at each finite pole a. Define*

$$N(f) = \max_{a \text{ a finite pole of } f} \left\lceil \frac{m_a}{e_a} \right\rceil, \qquad \max \varnothing = 0. \tag{6}$$

The following are equivalent:

1. *f is stable under d/dx in $\overline{\mathbb{C}(x)}$;*
2. *$f = \delta^N q$ for some $q \in \mathbb{C}[t]$, with $N = N(f)$;*
3. *f has an algebraic $N(f)$-fold primitive.*

In particular,

$$\mathrm{Stab}(\delta, \overline{\mathbb{C}(x)}) \cap \mathbb{C}(t) = \bigcup_{n\geq 0} \delta^n \mathbb{C}[t]. \tag{7}$$

For $f = 0$ all assertions hold trivially.

Proof. The implication (1)⇒(3) is immediate. For (3)⇒(2), Lemma 3.1 descends an algebraic N-fold primitive to some $q \in K$. Suppose q has a finite pole at a, of order $k > 0$. Lemma 3.2 implies that f has a pole there of order $m_a = k + Ne_a > Ne_a$. But the choice of N gives $Ne_a \geq m_a$, a contradiction. A pole of q outside the pole set of f is excluded by the same lemma. Hence q has no finite poles and is a polynomial. If $N = 0$, the hypothesis says simply that f has no finite poles, with the same conclusion.

For (2)⇒(1), polynomial primitives are supplied by J_P from (2). A coherent primitive sequence for $f = \delta^N q$ begins with

$$\delta^N q,\ \delta^{N-1} q, \ldots, q,\ J_P q,\ J_P^2 q, \ldots.$$

Every element lies in K. This proves the equivalence and the union description. The union is increasing: $\mathbb{C}[t] \subseteq \delta\mathbb{C}[t]$ because δJ_P is the identity on polynomials. □

Proposition 3.4 (A polynomial degree bound)**.** *Let $d = \deg P$ and, for a reduced fraction $f = A/B \neq 0$, put $\beta = \deg A - \deg B$. If $f = \delta^N q$ for a polynomial q, then*

$$\deg q \leq B_f := \max\{Nd + \beta,\ d(N-1)\}. \tag{8}$$

Here β is the growth degree at infinity, the negative of the valuation at infinity. For $N = 0$, the bound is $\deg q = \beta$.

Proof. Write $P(t) = ct^d + O(t^{d-1})$. If $M = \deg q > d(N-1)$, then successive leading terms under δ have degrees $M - d, M - 2d, \ldots, M - Nd$, and their combined leading coefficient contains the nonzero factor

$$(dc)^{-N} \prod_{j=0}^{N-1} (M - jd).$$

Consequently $\beta = M - Nd$ and $M = Nd + \beta$. If $M \leq d(N-1)$ there is nothing to prove. The case $N = 0$ is immediate. □

Theorems 3.3 and 3.4 give a terminating decision procedure over an effective coefficient field. Factor the reduced denominator of f, calculate its pole orders and the corresponding multiplicities in P', and form $N(f)$. Then substitute

$$q(t) = \sum_{j=0}^{B_f} c_j t^j$$

into $\delta^N q = f$, clear denominators, and solve the finite linear system. If $B_f < 0$, no nonzero polynomial candidate exists, so the answer is negative. A solution provides the whole primitive tower; inconsistency proves that no algebraic N-fold primitive, and hence no algebraic stability, is possible. Over $\mathbb{Q}$, factorization over $\mathbb{Q}$ suffices for computing N: all roots of a given irreducible factor have the same denominator and P' multiplicities.

This is a complete classification in the stated fixed polynomial cover. It does not assert that every algebraic function belongs to such a presentation.

4 Elementary stability and the scope of Problem 8

The distinction between algebraic and elementary primitives is essential. An algebraic function may fail the classification above and nevertheless admit arbitrarily many elementary primitives.

Lemma 4.1 (Polynomial moments and repeated integration)**.** *In a differential field with $Dx = 1$, if $H_j' = x^j f$ for all $j \geq 0$, then a coherent primitive sequence is given by $G_0 = f$ and*

$$G_n = \frac{1}{(n-1)!} \sum_{j=0}^{n-1} (-1)^j \binom{n-1}{j} x^{n-1-j} H_j, \qquad n \geq 1. \tag{9}$$

Conversely, the existence of all repeated primitives implies the integrability of all $x^j f$ in the same field.

Proof. Differentiating (9), the terms containing f cancel by the binomial theorem when $n > 1$, and the remaining terms give G_{n-1}. For $n = 1$, the derivative is f. Conversely, if $D^{j+1} g = f$, repeated integration by parts expresses a primitive of $x^j f$ as a finite sum of polynomial multiples of derivatives of g. □

This equivalence is already used in Chen–Kauers [1] and Chen [2, Lemma 2.10]; it is recalled to make the scope clear.

Proposition 4.2 (Rationally parametrized elementary stability)**.** *Let $x = R(t)$ for any nonconstant rational function $R \in \mathbb{C}(t)$. Every $f \in \mathbb{C}(t)$ is elementary-stable over $\mathbb{C}(x)$.*

Proof. For every $j \geq 0$, substitution gives

$$x^j f \, dx = R(t)^j f(t) R'(t) \, dt,$$

a rational differential in t. Its primitive is a rational function plus a finite constant linear combination of logarithms of rational functions in t, by partial fractions. Since t is algebraic over $\mathbb{C}(x)$, these primitives are elementary over that field. Lemma 4.1 constructs all repeated elementary primitives. □

For example, on $x = t^3 + t$, the function $f = 1/t$ has $N(f) = 1$ but no algebraic primitive:

$$\int f \, dx = \int \frac{3t^2 + 1}{t} \, dt = \frac{3}{2} t^2 + \log t.$$

Its nonzero simple residue prevents a rational primitive in $\mathbb{C}(t)$, and Lemma 3.1 prevents an algebraic primitive in any finite extension. Nevertheless it is elementary-stable by Proposition 4.2.

Chen–Kauers Problem 8 asks for the class of *all* elementary-stable functions. The results here settle neither higher-genus algebraic presentations in general nor arbitrary nested exponential and logarithmic extensions. They refute a particular proposed algebraic classification and give exact positive and negative results on a substantial family. The full original problem remains unresolved by this manuscript.

5 Concrete examples and exact verification

5.1 An elementary real branch

The cubic counterexample has a unique real branch, because $t \mapsto t^3 + t$ is strictly increasing. It can be written explicitly as

$$t(x) = \frac{2}{\sqrt{3}} \sinh\left(\frac{1}{3} \operatorname{arsinh}\left(\frac{3\sqrt{3}}{2} x \right) \right). \tag{10}$$

The identity $\sinh(3u) = 4\sinh^3 u + 3\sinh u$ verifies $t^3 + t = x$. Thus the example can also be presented using familiar elementary functions. Its exclusion from Chen's proposed class does not mean it is inexpressible using nested radicals or elementary compositions; it excludes the particular finite sum in (1).

Its first primitive is already enough to display the mechanism:

$$\frac{d}{dx}\left(\frac{3}{4}t(x)^4 + \frac{1}{2}t(x)^2\right) = \frac{3t^3 + t}{3t^2 + 1} = t.$$

The map J_P supplies every later primitive without introducing a larger field or an unproved limiting step.

5.2 Positive and negative classification examples

For $x = t^3 + t$, set $q(t) = 3t^2 + 1$. Representative outcomes of the finite algorithm are shown below. In this table "stable" means *algebraically* stable. All listed functions are elementary-stable by Proposition 4.2.

Input	$N(f)$	B_f	Algebraic outcome
t	0	1	Stable: polynomial tower
$1/q$	1	1	Stable: $\delta(t) = 1/q$
t/q	1	2	Stable: $\delta(t^2/2) = t/q$
$\delta^3(t^2)$	3	6	Stable: witness t^2
$1/t$	1	2	No algebraic first primitive
$1/q^2$	1	0	No algebraic first primitive

The second negative case follows because a candidate polynomial primitive would have degree at most zero and therefore derivative zero. This conclusion concerns primitives in the entire algebraic closure, by trace descent, not just a failed choice of rational ansatz.

On the cyclic cover $x = t^4$, the same algorithm accepts t^{-1} and t^{-7} but rejects t^{-4} and t^{-8}. For example,

$$\delta^2\left(-\frac{16}{3}t\right) = t^{-7}, \qquad t^{-8} = x^{-2}.$$

The latter has a first algebraic primitive $-x^{-1}$, followed by a logarithmic obstruction. This tests the distinction between one-time algebraic integrability and stability.

5.3 Verification record

The first embedded program uses only Python's exact rational arithmetic. It verifies the primitive recurrence for each $d = 3, \ldots, 10$ through depth 20: all 160 identities pass. It also checks the first three cubic primitives printed above and the degree of every tested tower element. These finite checks supplement the uniform proof, which applies to every $d \geq 3$ and every depth.

The second program implements the classification in Theorem 3.3 over $\mathbb{Q}$. It checks 14 examples, including ordinary poles, simple and multiple ramification, and delayed logarithmic obstructions. Every positive case returns a polynomial q and verifies $\delta^N q = f$ exactly. Every negative case returns a row vector w for the cleared linear system $Ac = b$ such that

$$wA = 0, \qquad wb \neq 0.$$

This is an exact inconsistency certificate. All 14 outcomes pass their independent identity or separating-functional checks. The proof of the degree bound is essential: an inconsistent arbitrary truncated ansatz would not certify nonexistence.

Reproduction from a single source. The source includes the bibliography and both Python programs in `filecontents*` blocks. Normal LaTeX compilation extracts the programs without executing them. Run

```
python3 verify_polynomial_inverse.py
python3 verify_cover_classifier.py
```

The first program uses the standard Python library; the second requires SymPy. Both produce JSON records. No shell escape, external bibliography, or separate figure file is required to compile the paper.

6 Consequences and remaining questions

The conjectured algebraic classification fails by explicit counterexamples of every degree at least three. The obstruction is structural: the proposed expression class lies in abelian extensions, whereas polynomial inverse branches need not generate normal extensions at all. Repeated algebraic integration is therefore compatible with this nonabelian field-theoretic behavior.

For polynomial covers, the corrected description (7) is complete and effective. It explains both the counterexamples and their stable rational derivatives, and gives finite negative certificates for functions that acquire logarithms after a bounded number of integrations. It does not propose a universal replacement classification for all algebraic functions.

The full Chen–Kauers Problem 8 still requires arbitrary elementary towers to be treated. This manuscript does not prove that every elementary-stable function is D-finite, does not classify positive-genus algebraic function fields in general, and does not certify the absence of earlier equivalent results. The counterexample proof is unconditional and self-contained; external review and a broader priority assessment remain separate tasks.

References

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